Project Proposal | 6COSC023W Final Year Project

On-Chain Signal Analysis for Crypto Trading

Testing whether blockchain transaction data predicts price and volatility beyond what market data alone shows

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Declaration

This project proposal has been prepared based on my own work. Where other published and unpublished source materials have been used, these have been acknowledged in references.

Tool	Purpose	How it was used
Claude (Anthropic)	Source summarising	11 reviewed sources were found and selected independently through Google Scholar and publisher pages. Due to the number of sources, AI was used to summarise their key findings.
Claude (Anthropic)	SPaG check + Structure framework	Proofreading (spelling, grammar, punctuation) and help structuring sentences
Claude (Anthropic)	Organising supervisor input	Asked to sort the resources listed in my supervisor's email into categories
Claude (Anthropic)	Explaining concepts	Asked to explain the statistical methods (e.g. walk-forward validation and the Diebold-Mariano test) so I could understand and apply them

Background

On-chain data is useful, but the tools are expensive

Retail crypto volume, Q1 2026

$979B

Retail transaction volume, down 11% from $1.1T a year earlier (TRM Labs, 2026)

Cost of an enterprise tool

~$50k-200k/yr

Chainalysis does not publish prices; one third-party estimate (CostBench, 2026)

This project's data cost

$0/mo

Free tiers only (BigQuery, The Graph, CCXT)

Stakeholders: individual traders (the users), paid platforms like Nansen, Glassnode and Chainalysis, and the free data sources the project uses. This project tests whether this kind of information can be checked and given to an individual trader for free.

Problem and Motivation

Chart-only traders may miss signals that appear before the price moves

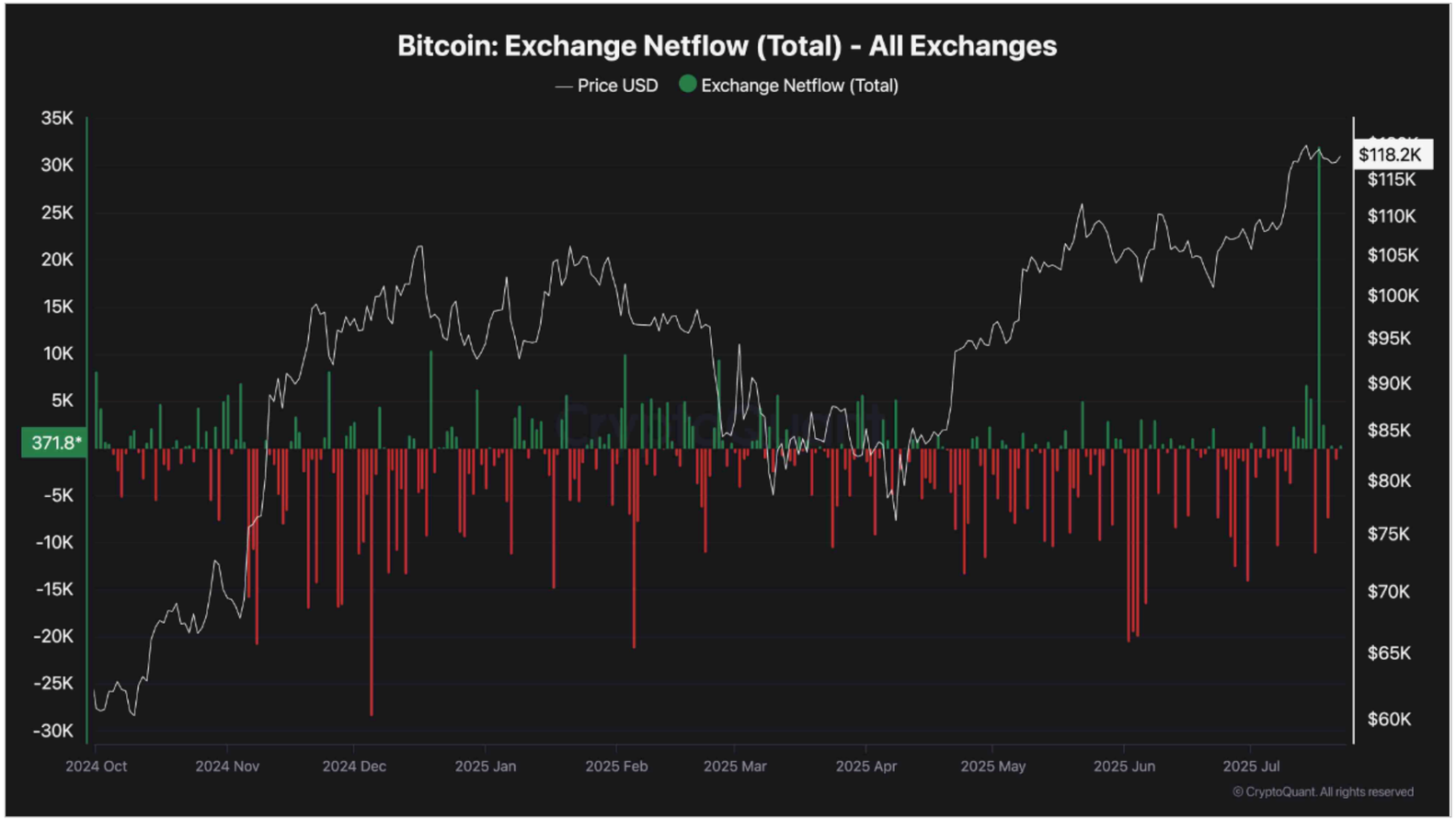
Traders who rely only on price charts may miss a category of information that can appear before the price moves: on-chain activity. Some studies find that on-chain data helps forecast returns or market cycles (Chi, Chu and Hao, 2024; Grobys, Näsman and Sandretto, 2026), though the evidence is mixed.

Wallet moves funds to an exchange → **Chart-only trader sees nothing yet** → **Price drops: they notice too late**

This project asks: do on-chain signals carry predictive information beyond market data, and can a free tool give an individual trader this early visibility?

The Idea

The kind of signal this project tests: Bitcoin exchange flows vs price



Illustrative only: CryptoQuant data is not part of this project's pipeline, which builds the same net-flow measure from BigQuery and exchange wallet labels. Bars: daily net flow into (green) or out of (red) exchanges; line: price. Spikes like July 2025's have often come before corrections (Shayan Markets, 2025), but research is mixed: ETH inflows predict lower ETH returns, while BTC inflows mostly do not (Chi, Chu and Hao, 2024). Objective 4 tests whether these flows improve the forecast; Objective 5 measures how far ahead they appear.

Literature Review

11 sources: mixed evidence, and none combines all four signals

Do signals predict price?

Chi, Chu and Hao (2024): ETH and USDT flows predict returns at 1-6 hours; BTC flows mostly do not [Obj 4-5]

Grobys, Näsman and Sandretto (2026): on-chain data predicts BTC cycles [Obj 6]

Chalkiadakis et al. (2022): statistical causality, sentiment & price (single signal) [Obj 4]

Signal data and regimes

Drakopoulou (2026): stablecoin liquidity as a regime signal [Obj 2]

Ante, Fiedler and Strehle (2021): stablecoin issuance and abnormal crypto returns [Obj 2]

Zhu et al. (2024): what drives DEX liquidity on Uniswap [Obj 2]

Urumov and Chountas (2022): fuzzy clustering for volatility (supervisor's paper) [regimes]

Entities, methods, validation

Kang et al. (2020): de-anonymisation via address clustering [Obj 2]

Granger (1969): supporting causality test in Objective 4 [Obj 4]

Bailey et al. (2016): backtest overfitting in finance [Obj 7]

Harvey, Liu and Zhu (2016): multiple-testing bias in finance research [Obj 7]

Gap: none of the reviewed studies combines all four signal types with this validation design

Bailey et al. (2016) and Harvey, Liu and Zhu (2016) warn that testing many signals risks false discovery, hence walk-forward validation (Objective 7).

Trend: newer studies (2026) look at market cycles, while older ones test one signal at a time. Results change with the time horizon and the test used (Chi, Chu and Hao use hours; Drakopoulou uses days), so this project fixes its two targets, four lead-lag windows and walk-forward test in advance.

Possible extension: test each signal inside regimes found by fuzzy clustering (Urumov and Chountas, 2022).

Existing Solutions and Requirements

Existing platforms offer on-chain analytics; this project tests four signals combined.

Platform	Pricing	What it does	Gap it leaves
Nansen	Free, or Pro from $49/mo	Wallet labelling	No published predictive testing
Glassnode	Paid plans from $49/mo	Macro on-chain metrics	Predictive testing only in beta (Alpha Lab)
Arkham Intelligence	Free to browse; API needs approval	Entity labelling (planned label source)	No forecasting or signal testing in its documented features
Chainalysis	Quote-based (est. $50k-200k/yr)	Blockchain forensics (compliance)	Priced for institutions, not individuals

Requirements drawn from these gaps

R1
Combine all 4 signals into one tested score
none of the four platforms reviewed publishes one

R2
Test each signal statistically, not just display it
forecast tests and walk-forward validation

R3
Keep data free
free tiers only, so any individual can use it (SDG 10)

R4
Show confidence ranges and hit rate, not buy/sell advice
limits misuse and over-trust (SDG 10)

The project does not collect new data. It adds the testing that these platforms do not provide.

Sources: Nansen (2026), Glassnode (2026), Arkham Intelligence (2026a, 2026b), CostBench (2026).

Aim

Develop a web dashboard for traders that surfaces on-chain signals across Bitcoin and Ethereum, to evaluate whether they predict price movement and volatility beyond market data alone.

Exchange inflows/outflows	**Whale wallet transfers**	**Stablecoin flows**	**DEX liquidity activity**

Research question: does each signal add predictive information beyond market data, and does combining all four beat the best single signal?

Why it matters (SDG 10): checking on-chain data should not need a large budget.

Objectives

7 SMART objectives, each with a pass or fail test

1. **20 Nov 2026** Pipeline for BTC/ETH (BigQuery, CCXT): 3 months of history at each source's finest resolution (5-minute for prices), under 5% missing intervals, plus a live feed (Alchemy/Infura, Mempool.space)
2. **8 Jan 2027** Collect the 4 signals via BigQuery, exchange wallet labels and The Graph: 8 series (4 signals, 2 assets), each covering the full history
3. **8 Jan 2027** Rolling z-score anomaly detection. Success: flags transfers over 2 standard deviations from that wallet/token's recent mean
4. **21 Jan 2027** Market-only model (past returns and trading volume) vs the same model plus each signal, on a walk-forward split. Success: lower out-of-sample RMSE for next-hour log returns and realised volatility (two pre-registered targets), significant on a Diebold-Mariano test after Holm correction; Granger is supporting evidence only
5. **21 Jan 2027** Lead-lag: correlation of each signal with later returns and volatility at 30min, 1hr, 4hr and 24hr. Success: the shortest window that is significant after Holm correction and holds out-of-sample is the lead time; otherwise no lead is claimed
6. **10 Mar 2027** Composite score of all 4 signals. Success: beats the best single signal (chosen on training data only) out-of-sample on a Diebold-Mariano test after Holm correction; otherwise a negative result
7. **10 Mar 2027** Walk-forward validation across objectives 4-6: expanding window, 9 weekly test folds scored as one pooled test. Success: a relationship counts as confirmed only if it holds out-of-sample

Realistic: each objective can be tested, simplified or dropped independently, within the module's 400-hour budget.

Scope

What this project does and does not cover

Included

- Data pipeline for Bitcoin and Ethereum (Objective 1)
- The 4 signal types and anomaly detection (Objectives 2-3), plus cross-exchange and cross-chain flows from BigQuery (Q7)
- Statistical tests: forecast comparison (Diebold-Mariano), lead-lag, composite score, walk-forward validation (Objectives 4-7)
- Everything delivered in one web dashboard

Excluded

- Cryptocurrencies beyond Bitcoin and Ethereum (keeps 7 objectives achievable in the 400-hour budget)
- Live trading and profit claims (the project tests prediction; fees and slippage are not modelled)
- Claims beyond this sample (results hold for BTC and ETH over these 3 months only)
- A mobile app (web only, built with Streamlit)
- Any on-chain signals beyond the 4 chosen types (each extra signal adds its own data pipeline and tests)

System Overview

From raw blockchain data to a tested dashboard

Sources
Raw data

BigQuery
on-chain SQL history

Arkham
wallet labels, if approved

The Graph
DEX liquidity

CCXT
market prices

Alchemy + Mempool.space
live feed

→

Objective 1
Pipeline

Python + SQLite
one stored dataset for BTC and ETH

3 months history
plus a live feed

→

Objective 2
Four signals

Exchange flows

Whale transfers

Stablecoin flows

DEX liquidity

→

Objectives 3-7
Testing

3 Anomaly flags
rolling z-score

4 Added value
Baseline vs +signal

5 Lead time
signal vs later returns and volatility

6 Composite
4 signals combined

7 Walk-forward
out-of-sample

→

Output
Dashboard

Streamlit app
BTC and ETH, free stack

The Deliverable

What the dashboard will show a trader

- The 4 signals for Bitcoin and Ethereum, with unusual moves flagged (Objective 3)
- A price chart with the flagged moves marked on it
- Whether each signal improves the forecast of returns and volatility beyond market data, with its p-value (Objective 4)
- The lead time of each signal, up to 24hr (Objective 5)
- A combined score with a range (Objective 6)
- The hit rate on unseen data, and a status: confirmed, not confirmed or negative result (Objective 7). Returns: share of test hours where the predicted direction (up or down) was right. Volatility has no direction, so it shows forecast error against the baseline instead

Methodology

Agile (solo-adapted), with a success metric for each objective

Why Agile: each objective is tested independently, so any can be dropped without stalling the others; Waterfall would lock in the composite score before any signal is shown to work. **Limitation:** Agile assumes a team and flexible deadlines; this is a solo project with fixed module deadlines, so each sprint is tied to a phase date in the Time Schedule.

Requirements elicited from: supervisor consultation (14 research questions, feedback on data and tests) and the literature gap analysis (R1-R4).

Objective	Method	Success metric	Limitation
1-2 · Data and signals	BigQuery SQL and API pulls, stored in SQLite	Under 5% missing intervals	BigQuery lags the chain tip
3 · Anomaly detection	Rolling z-score per wallet/token	Flags transfers over 2 SD	Assumes bell-shaped data; crypto has fat tails
4 · Beyond-market test	Past returns and volume model vs same model plus signal; ADF test first; Granger as support	Lower out-of-sample RMSE (log returns, realised volatility); Diebold-Mariano $p < 0.05$ after Holm (16 tests)	Shows prediction, not cause
5 · Lead-lag	Signal vs later returns and volatility at 4 windows	Shortest window significant after Holm (64 tests)	Linear only; depends on the windows chosen
6 · Composite score	Equal-weight mix of the 4 z-scored signals, fixed in advance	Beats the best single signal (picked on training data); Diebold-Mariano after Holm (4 tests)	Equal weights may not be optimal; fitted weights only a side check
7 · Walk-forward validation	Expanding window (4-week minimum), rolled weekly; no look-ahead	Pooled RMSE over 9 test weeks (about 1,500 hours)	Less data per test; 3 months may miss other market conditions

Original contribution: the methods are standard, but no reviewed study tests all four signals together for Bitcoin and Ethereum with walk-forward validation.

Methodology in detail

Test design: every rule is set before testing starts

Part	Rule
Targets	Next-hour log return, and next-hour realised volatility (square root of the sum of squared 5-minute returns in that hour)
Baseline (Obj 4)	Market-only baseline: OLS on the target's own last 24 hourly values plus the last 24 hourly trading volumes (log), refitted in each training window (one fixed baseline, not every possible market model)
Signal model (Obj 4)	The same baseline, unchanged, plus one on-chain signal's last 24 hourly values
Decision (Obj 4)	Lower pooled out-of-sample RMSE, confirmed by a Diebold-Mariano test (Diebold and Mariano, 1995) with Holm correction over 16 tests (Holm, 1979)
Lead-lag (Obj 5)	Correlation with returns and volatility over the next 30min, 1hr, 4hr and 24hr; Newey-West errors, as overlapping windows are autocorrelated (Newey and West, 1987); Holm over 64 tests
Timing	A transfer counts in the hour its block is mined and is used only from the next hour; a daily signal uses its last completed day
Composite (Obj 6)	Equal weights on z-scored signals, fixed in this proposal; fitted weights are a side check only. The best single signal is chosen on training data only, then both are compared on the same test weeks by Diebold-Mariano, with Holm over 4 tests (2 assets × 2 targets)
What a pass means	Predictive value for BTC and ETH in this 3-month sample only, not profit: fees and slippage are not modelled, so the dashboard reports hit rate, not returns

Methodology in practice

Walk-forward validation: expanding window, no look-ahead

Walk-forward validation (expanding window)

Week of the 3-month sample

Fold	Train weeks	Test week	1	2	3	4	5	6	7	8	9	10	11	12	13
1	1-4	5													
2	1-5	6													
3	1-6	7													
4	1-7	8													
5	1-8	9													
6	1-9	10													
7	1-10	11													
8	1-11	12													
9	1-12	13													

Key

Training weeks (model fitted here)

Test week (model fixed before this week)

Each fold keeps all earlier weeks (4-week minimum). 9 test weeks x 168 hours = about 1,500 out-of-sample hours, scored as one pooled RMSE.

- **Set-up:** 3 months (13 weeks); first test is week 5, then roll weekly: 9 test weeks (about 1,500 out-of-sample hours), scored as one pooled RMSE; per-week wins are a stability note only
- **No look-ahead:** scaling is fitted inside each training window, and each signal uses only data known at that hour
- **Why it matters:** it stops lucky results looking real (Bailey et al., 2016; Harvey, Liu and Zhu, 2016). Other studies use years of data (Chi, Chu and Hao, 2024; Drakopoulou, 2026), so a negative result is still valid, and claims cover this sample only

Tools and Techniques

A free data stack, with alternatives considered

Need	Chosen	Why chosen over the alternative	Main risk
On-chain history (Obj 1-2)	**Google BigQuery public datasets**	Dune: free plan is view-only. Etherscan: lookups only, kept as secondary	Lags the chain tip; first 1 TiB/month free (Google Cloud, 2026)
Live feed (Obj 1)	**Alchemy / Infura (ETH), Mempool.space (BTC)**	BigQuery: lags the chain tip	Free-tier rate limits
Wallet labels (Obj 2)	**Arkham Intelligence** (API needs approval)	Nansen: free tier limited, Pro is paid. Whale Alert: no free history	Access not yet confirmed; fallback: exchange-published addresses
DEX liquidity (Obj 2)	**The Graph**	Glassnode: full depth is paid	Free query quota
Market prices (Obj 1)	**CCXT**	Direct exchange APIs: one integration each	Rate limits
Storage (Obj 1)	**Python, pandas, SQLite**	PostgreSQL: not needed for one user	Will not scale to many users
Statistics (Obj 3-7)	**statsmodels, scipy**	R: splits the codebase in two	Granger needs stable series
Dashboard (output)	**Streamlit**	Flask: charts must be built by hand	Reruns on every click

Hardware: a Mac laptop for coding and a Windows laptop for planning and writing; no GPU or paid cloud needed. **Development:** Python, VS Code, Git and GitHub. The dashboard runs locally and can be deployed online later if time allows. **Skills:** existing: Python, SQL, Git, Flask. To learn: BigQuery, The Graph queries, time-series testing (Diebold-Mariano, Newey-West) and Streamlit.

Data feasibility

Where each signal comes from, and the fallback

Data	Source	Resolution	Fallback if it fails
Exchange flows	BigQuery BTC and ETH tables, plus Arkham exchange labels (not yet confirmed)	Every block (BTC about 10 min, ETH 12 s)	Addresses exchanges publish, e.g. Binance (Quinn, 2022); main wallets only, so flows are partial
Whale transfers	BigQuery transfers, plus Arkham labels if approved	Every block	Size threshold only, without labels
Stablecoin flows	BigQuery Ethereum token transfers (USDT, USDC)	Every block	Etherscan token-transfer lookups
DEX liquidity	The Graph (Uniswap subgraph)	Hourly pool data	Alchemy: read pool events directly
Cross-chain flows (Q7)	BigQuery Ethereum transfers into and out of bridge contracts, and WBTC mints and burns	Every block	Etherscan lookups on the bridge contracts
Prices (targets)	CCXT (Binance)	5-minute candles	Another exchange through CCXT
Live feed	Alchemy/Infura (ETH), Mempool.space (BTC)	Seconds	BigQuery, accepting its lag

Checked 13 Oct 2026 (free sandbox): a test query on the BigQuery Bitcoin and Ethereum tables found them up to date, with the newest blocks about 30 min (BTC) and 1 min (ETH) behind the live chain. A 3-month pull of Ethereum token transfers would scan 47 GB, under 5% of the free 1 TiB a month.

Not yet confirmed: Arkham's API needs access approved by Arkham, and its terms limit how its data is used and shared (Arkham Intelligence, 2026a), so label access and terms are checked in week 1 before any labels are stored.

Go/no-go: every source is tested in week 1 of the pipeline phase; a signal whose source still fails is dropped and reported, and the other signals continue (Objective 2).

Ethics, Legal, Social, EDI and Sustainability

Key risks and safeguards

Area	Issue	Safeguard
Legal	Each source has its own terms; Arkham bans scraping and republishing its data	Follow each API's terms: approved access only (Computer Misuse Act 1990), no republishing (Copyright, Designs and Patents Act 1988)
Privacy (UK GDPR, Data Protection Act 2018)	Wallet addresses may count as "online identifiers"; clustering could link them to people	Only public entity labels (exchanges, funds), never individuals; no de-anonymisation
Data handling	Stored data could outlive its purpose	Public data only; backed up to University OneDrive; raw data kept off GitHub and deleted after final marks
Social and financial harm	Predictions misread as advice; crowd use could amplify volatility	Disclaimer, plus hit rate, range and status on every output; no buy or sell signals
EDI	Stablecoin users in unstable economies are often non-experts	Plain-language labels; no "get rich" marketing
Data and result quality	Data gaps, mislabelled wallets, false positives	Under 5% missing data; walk-forward validation; negative results reported
Sustainability	Cloud queries and model re-runs use energy (no mining or transactions)	Store data locally; filter BigQuery scans by date and column

Main goal, SDG 10 (target 10.2): the project runs on free data and a normal laptop, so it costs $0, compared with an estimated $50k-200k a year for enterprise tools (R3). Every output shows its hit rate, so users can judge how far to trust it (R4).

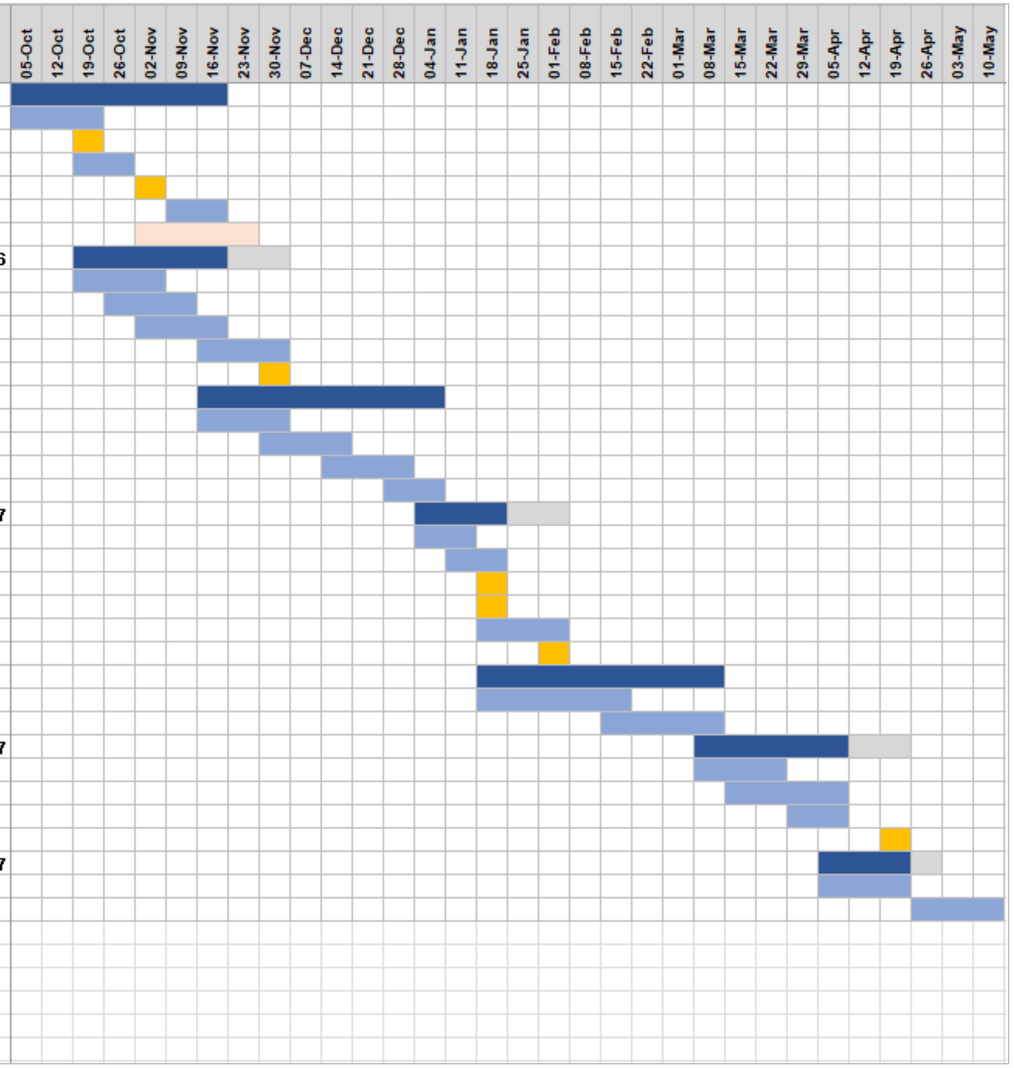
Supporting, SDG 8 (target 8.2, innovation): free analytics built on public blockchain data.

Limit: SDG 8.10 is about access to financial services; this widens access to analysis only. About 1.3 billion adults had no account in 2024 (Klapper et al., 2025).

Time Schedule

A ~2-week buffer ahead of every summative submission

Task	Start	End	Buffer to
1 PP finalisation	05/10/26	19/11/26	
1.1 Draft slides and supervisor review	05/10/26	21/10/26	
1.2 Formative draft submitted	22/10/26	22/10/26	
1.3 Supervisor corrections, ethics declaration	23/10/26	01/11/26	
1.4 PP submission (slides + ethics form)	02/11/26	02/11/26	
1.5 Live presentation window (requested online)	09/11/26	19/11/26	
Away 2–23 Nov	02/11/26	23/11/26	
2 Data pipeline	21/10/26	20/11/26	04/12/26
2.1 BigQuery queries (date- and column-filtered)	21/10/26	03/11/26	
2.2 Live feed: Alchemy/Infura (ETH), Mempool.space (BTC)	28/10/26	10/11/26	
2.3 CCXT prices, SQLite schema, 3-month backfill	04/11/26	20/11/26	
2.4 Systems Requirement Specification (SRS)	21/11/26	03/12/26	
2.5 SRS deadline	04/12/26	04/12/26	
3 Signals + anomaly detection	21/11/26	08/01/27	
3.1 Exchange flows (label go/no-go check)	21/11/26	04/12/26	
3.2 Whale transfers (Arkham labels if approved)	05/12/26	18/12/26	
3.3 Stablecoin flows (BigQuery) and DEX liquidity (The Graph)	19/12/26	01/01/27	
3.4 Rolling z-score anomaly detection	02/01/27	08/01/27	
4 Statistical testing + MVP	09/01/27	21/01/27	04/02/27
4.1 Baseline vs baseline + signal, Diebold-Mariano, Holm (16 tests)	09/01/27	15/01/27	
4.2 Lead-lag at 30min, 1hr, 4hr, 24hr (64 tests)	13/01/27	21/01/27	
4.3 First code demo	22/01/27	22/01/27	
4.4 Ethics declaration accepted (required by IPD)	22/01/27	22/01/27	
4.5 IPD pack (slides, prototype, video)	23/01/27	03/02/27	
4.6 IPD deadline	04/02/27	04/02/27	
5 Composite score + validation	22/01/27	10/03/27	
5.1 Composite score (fixed equal weights), Holm (4 tests)	22/01/27	15/02/27	
5.2 Expanding walk-forward, 9 test weeks, pooled RMSE	16/02/27	10/03/27	
6 Full dashboard + report	11/03/27	08/04/27	22/04/27
6.1 Polish dashboard UI	11/03/27	24/03/27	
6.2 Write report (methods, results, reflection)	18/03/27	08/04/27	
6.3 Record demo video	01/04/27	08/04/27	
6.4 Final submission (report, code, video)	22/04/27	22/04/27	
7 Viva prep	09/04/27	20/04/27	28/04/27
7.1 Mock Q&A across all 7 objectives	09/04/27	20/04/27	
7.2 Viva window	29/04/27	14/05/27	
Phase			
Task			
Buffer before deadline			
Milestone / deadline			
Away (talk requested online)			



Official deadlines: PP 2 Nov, SRS 4 Dec, first code demo 22 Jan, IPD 4 Feb, final report/code/video 22 Apr, viva 29 Apr-14 May.

Time Schedule detail

Seven phases, each with a scored risk and a fix

1 PP finalisation

Draft 22 Oct (formative); deadline 2 Nov: slides and ethics form; talk 9-19 Nov

Risk (5×2 = 10, medium): Away 2-23 Nov, during the talk window

Fix: Talk requested online; rehearsal done before leaving

2 Data pipeline

Target 20 Nov: BigQuery, live APIs, SQLite, 3-month backfill; SRS by 4 Dec

Risk (2×4 = 8, medium): BigQuery 1 TiB free quota and rate limits

Fix: Go/no-go check of each source in week 1; filter scans by date and column; coarser data if needed

3 Signals + anomaly detection

Target 8 Jan: 4 signals, rolling z-score

Risk (4×3 = 12, medium): Arkham access refused, or labels harder than expected

Fix: Exchange-published addresses; if a signal is still unusable after 2 weeks, drop it and report on three

4 Statistical testing + MVP

Target 21 Jan: forecast tests, lead-lag, first dashboard; IPD pack by 4 Feb, with the ethics declaration accepted by then. Needs phases 2 and 3

Risk (4×2 = 8, medium): No significant result

Fix: Report as a negative result; show behaviour descriptively

5 Composite score + validation

Target 10 Mar: composite (fixed equal weights), walk-forward test. Needs phase 4

Risk (2×2 = 4, low): Backtesting compute time

Fix: Start simple before optimising

6 Full dashboard + report

Deadline 22 Apr: polish UI, write results, demo video

Risk (3×4 = 12, medium): Report-writing crunch

Fix: Draft each section as its objective completes

7 Viva prep

Before 29 Apr: mock Q&A across all 7 objectives

Risk (3×4 = 12, medium): Statistics questions that are hard to answer

Fix: Mock Q&A with supervisor feedback; ~1-week buffer

Risk scoring: probability × impact, each 1-5 (15+ high, 6-14 medium, under 6 low). **Strategy:** ~2-week buffers before summative submissions; risks reviewed at each supervisor meeting.

Reflection: a short log after each phase, reused in the final report.

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